

Swinburne University of Technology
School of Science, Computing, and Emerging Technologies

SWE40006 - Software Deployment and Evolution
Deployment Portfolio Task 1

Student name: Ngoc Van Nhi Do
Student ID: 105551765

Deployment Portfolio Task 1

1 Declared Target Level

All four subtasks 1.1, 1.2, 1.3, and 1.4 were attempted.

2 Execution Evidence

2.1 Task 1.1

I built and packaged a sample console application into an .msi installer using WiX 7.0.0 via the Command-line Interface (CLI).

The first step is creating a new C# project, adding the provided code, and building it in Release mode.

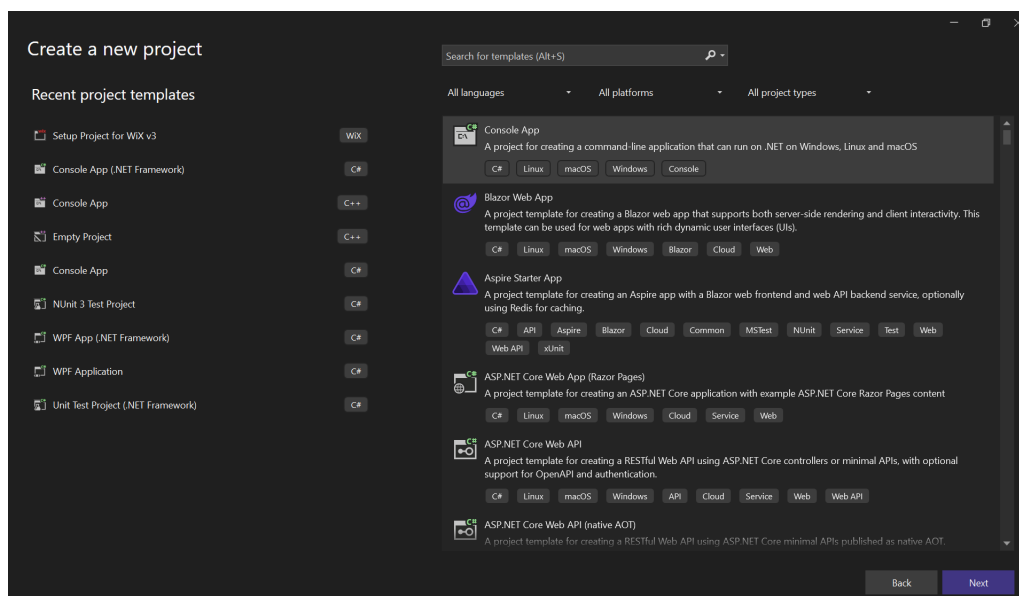


Figure 1: Creating a new C# console application.

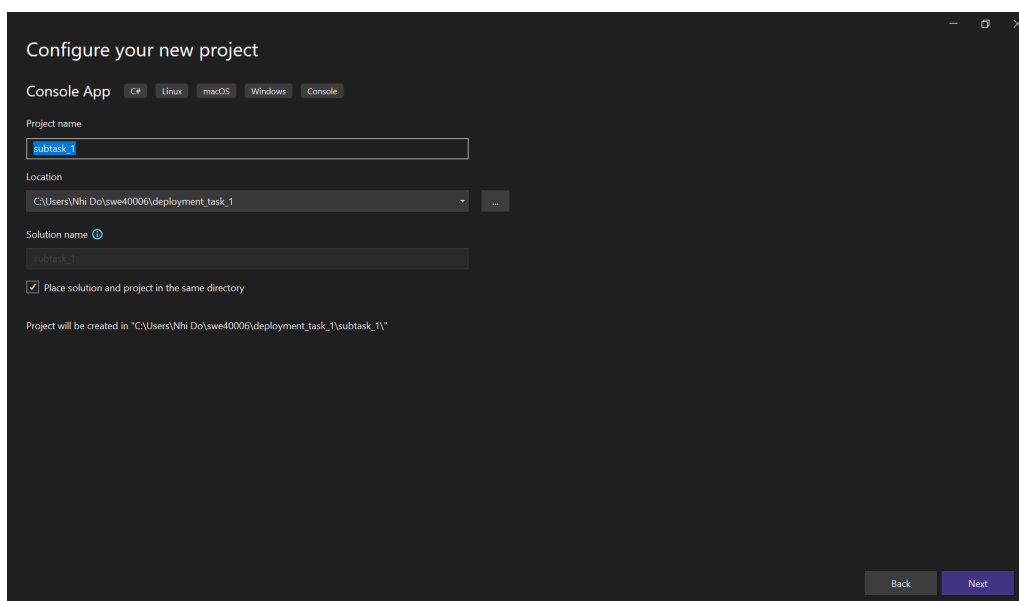


Figure 2: Naming the application.

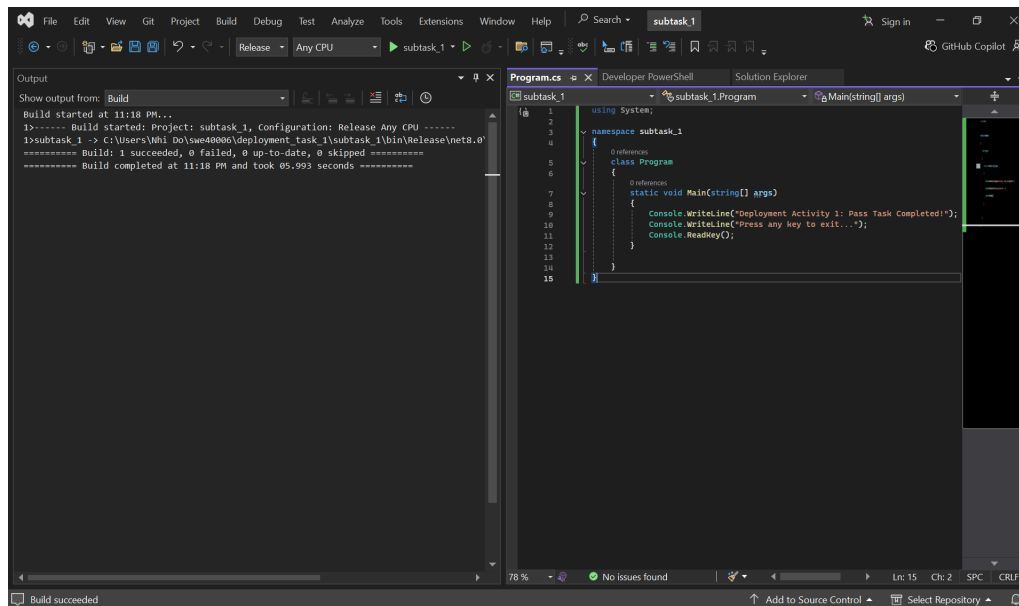


Figure 3: Building the program in Release mode.

Once the program has been built, its executable is created the project's **bin Release** folder.

The WiX toolset is installed using the CLI.

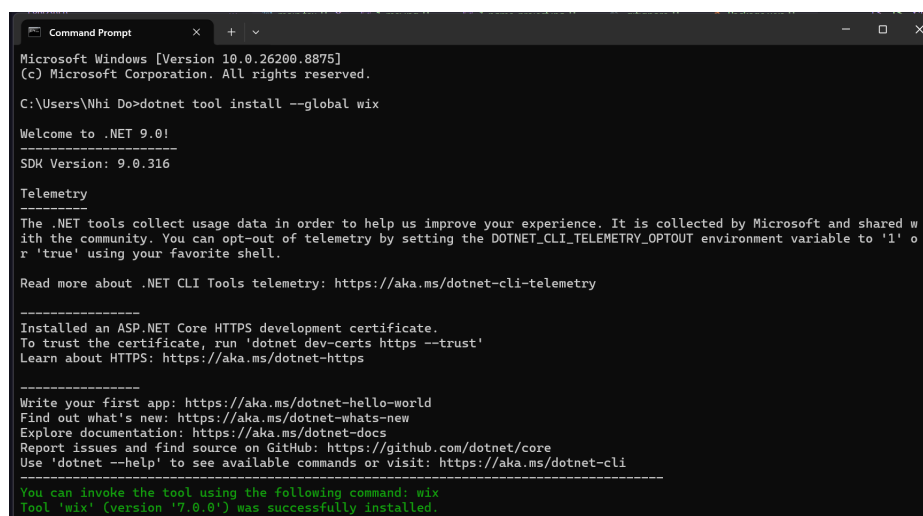


Figure 4: Installing WiX.

The WiX project and source file are manually created. The **.wxs** file is then configured to contain instructions to build the installer. The file specifies the application name, manufacturer, version, and upgrade code, as well as the directory in which the application will be installed. It also defines the application files that should be included in the installer and groups them into the feature.

```

File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help Search subtask_1
Release Any CPU subtask_1
Program.cs Developer PowerShell Solution Explorer
+ Developer PowerShell
PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1> mkdir wix_installer

Directory: C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1

Mode                LastWriteTime         Length Name
----                -
d-----            8/8/2026 11:26 PM             wix_installer

PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1> cd .\wix_installer\
PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer> type nul > Package.wxs
type : cannot find path 'C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer\nul' because it does not exist.
At line:1 char:1
+ type nul > Package.wxs
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\Nhi Do\...x_installer\nul:String) [Get-Content], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer>

```

Figure 5: Creating the WiX installer project and source file.

```

File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help Search subtask_1
Release Any CPU subtask_1
Package.wxs Program.cs Developer PowerShell
1 <?xml xmlns="http://wixtoolset.org/schemas/vd/wxs">
2
3 <Package
4   Name="Subtask 1"
5   Manufacturer="Sainmy Student"
6   Version="1.0.0"
7   UpgradeCode="1fcb3dfc-9388-4898-b478-07c0b69623c1">
8
9   <MajorUpgrade
10    DowngradeErrorMessage="A newer version of Subtask 1 is already installed." />
11
12   <MediaTemplate />
13
14   <StandardDirectory Id="ProgramFilesFolder">
15     <Directory Id="INSTALLFOLDER" Name="subtask_1" />
16   </StandardDirectory>
17
18   <Feature Id="MainFeature" Title="subtask_1" Level="1">
19     <ComponentGroupRef Id="AppFiles" />
20   </Feature>
21
22 </Package>
23
24 <Fragment>
25   <ComponentGroup Id="AppFiles" Directory="INSTALLFOLDER">
26     <File Source="..\bin\Release\net8.0\subtask_1.exe" />
27     <File Source="..\bin\Release\net8.0\subtask_1.dll" />
28     <File Source="..\bin\Release\net8.0\subtask_1.deps.json" />
29     <File Source="..\bin\Release\net8.0\subtask_1.runtimeconfig.json" />
30   </ComponentGroup>
31 </Fragment>
32
33 </Wix>

```

Figure 6: Configuring the Package.wxs file to build the installer.

The installer is then built using the command below. The resulting MSI file is created inside the installer directory.

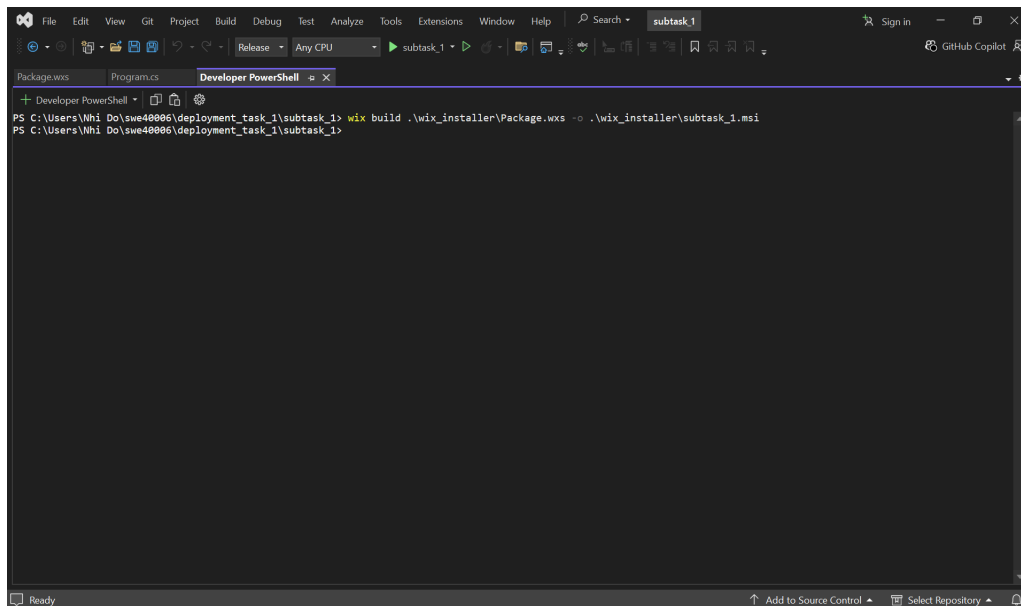


Figure 7: Building the installer from the WiX source file.

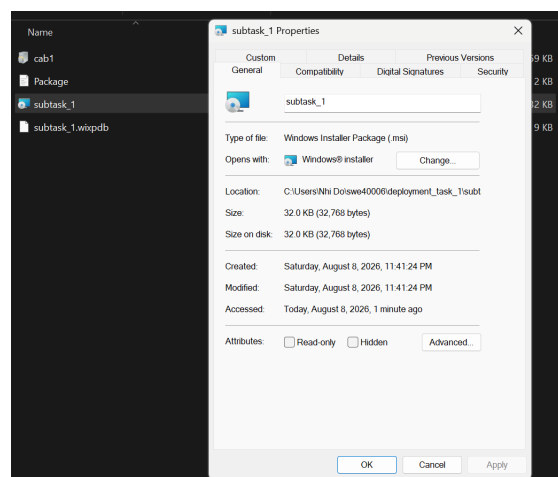


Figure 8: The MSI file is created.

Running the installer `subtask_1.msi` installs the application. The screenshot below shows that the app is successfully installed in the Program Files(x86) folder and runs as expected.

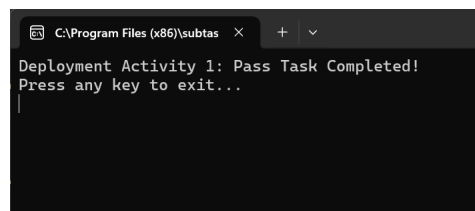


Figure 9: Running the console application after it has been installed.

The application Subtask 1 is finally uninstalled.

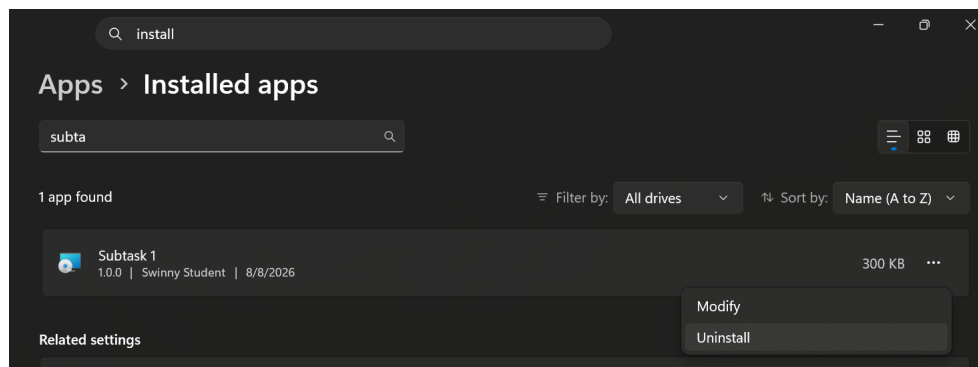


Figure 10: Uninstalling the application.

2.2 Task 1.2

For task 1.2, I built and packaged a C++ console grading application also using the WiX toolset.

First, I created the program and built it in Release mode. The resulting executable code is created in the `x64` directory.

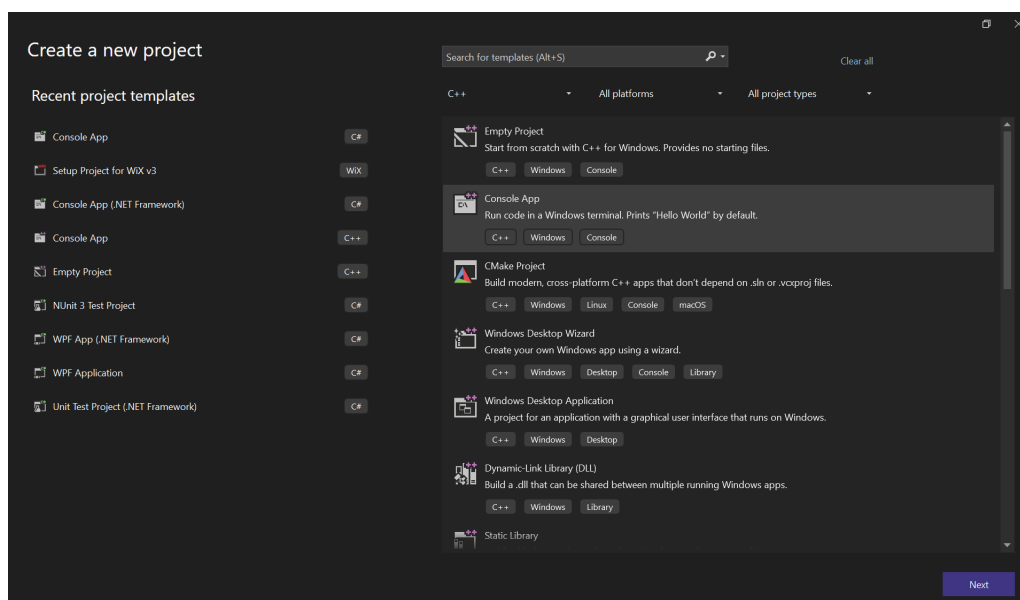


Figure 11: Creating a new C++ console application.

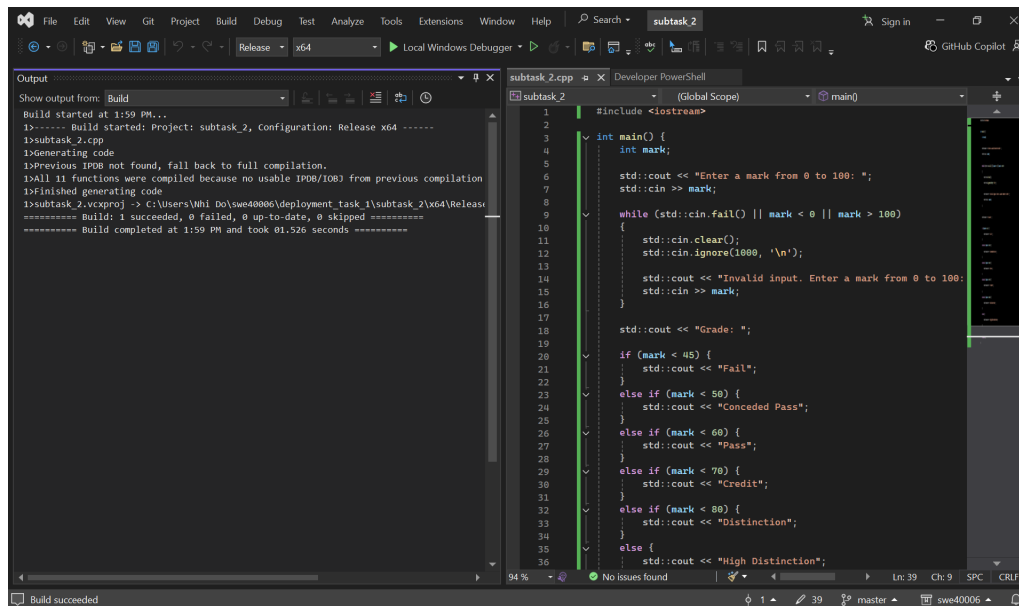


Figure 12: Building the project in Release mode.

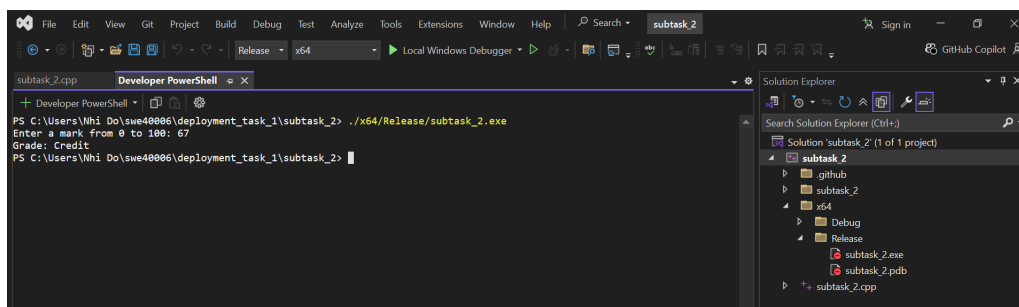


Figure 13: Verifying that the executable exists and runs correctly. The executable is created in the x64/Release directory as seen in the Solution Explorer.

Similarly, a WiX installer application is created, and a .wxs file is written to configure the installer.

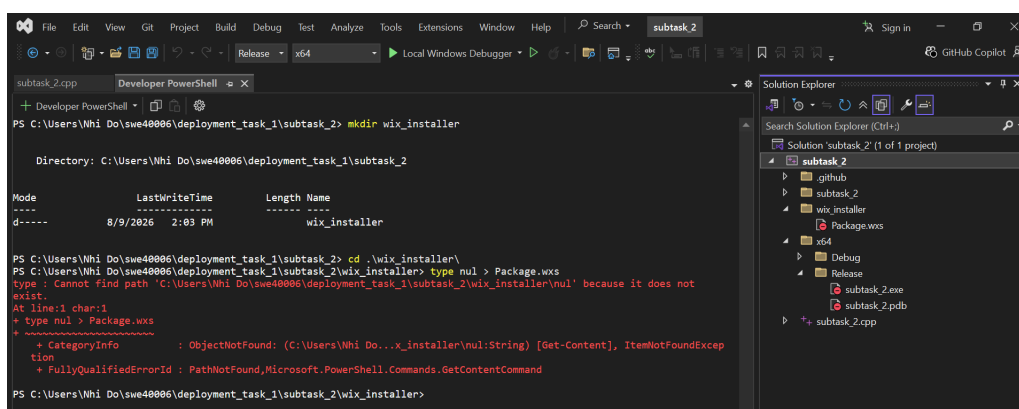


Figure 14: Creating the WiX installer project and source file.

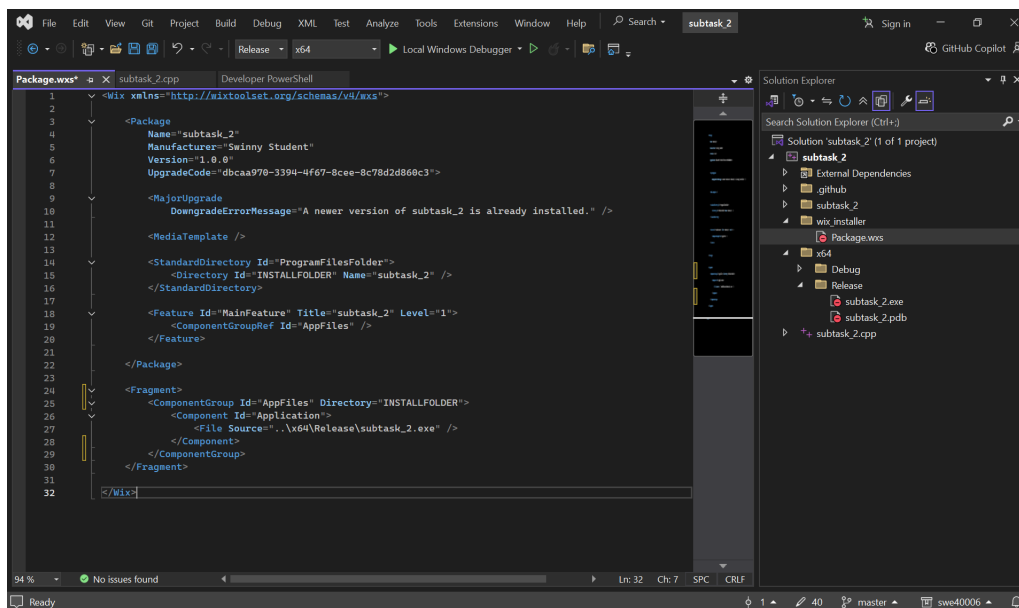


Figure 15: Writing the .wxs file.

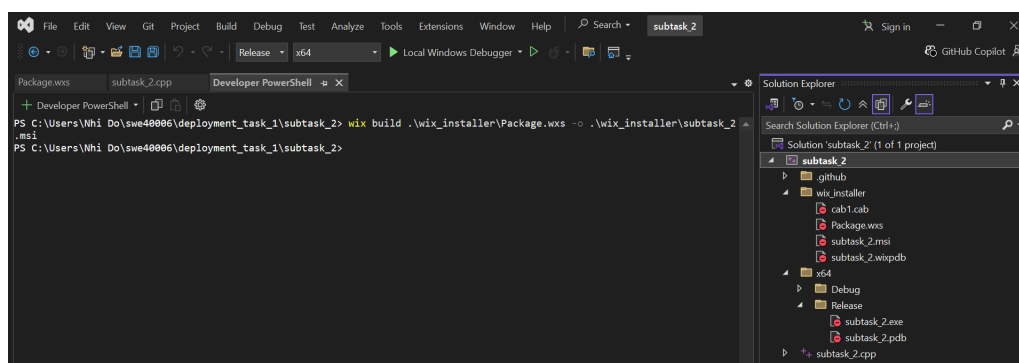


Figure 16: Building the MSI package.

Running the MSI package installs the app in the Program Files(x86) folder. The application runs as intended.

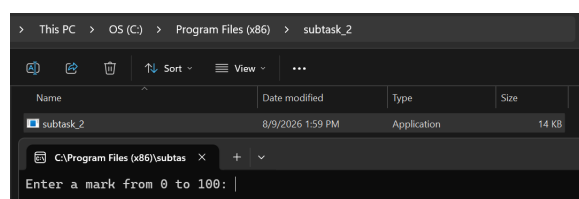


Figure 17: The application is installed in the Program Files (x86) folder. Double-clicking the app executes the program.

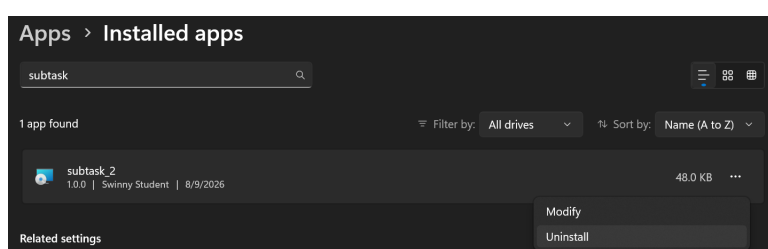


Figure 18: Uninstalling the application.